

CRANFIELD UNIVERSITY

School of Aerospace, Transport and Manufacturing

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# CFD Simulation of Water Flow through a Variable Area Channel

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Computational Engineering Fluids (B25)

Module Lead: Prof. Karl W. Jenkins

**Student Name:** John HOARAU

**Student ID:** 484338

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**Declaration of AI Tool Use:**

I acknowledge the use of generative Artificial Intelligence (Gemini 3.1 Pro) in the preparation of this assignment. Its use was restricted to proofreading, grammar correction, and structural refinement. All technical work, simulations, calculations, analysis, and conclusions presented in this report are entirely my own original work, in accordance with Cranfield University's academic integrity guidelines.

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### Abstract

This report presents a computational fluid dynamics (CFD) investigation into the flow of water through a two-dimensional rectangular channel containing a bluff body (a plate attached to the upper wall). Three volumetric flow rates (4 L/s, 8 L/s, and 12 L/s) are simulated using ANSYS Fluent 2024 R2 with the SST  $k-\omega$  turbulence model to analyse pressure drop characteristics and flow field behaviour. Area-weighted average pressures are extracted at measurement lines positioned 25 mm upstream ( $x = 0.205$  m) and 25 mm downstream ( $x = 0.285$  m) of the bluff body. The resulting pressure differences are 83.60 Pa, 333.22 Pa, and 745.42 Pa for the three flow rates respectively, demonstrating the expected quadratic relationship between pressure drop and flow velocity. A grid independence study comparing coarse (13 580 cells) and fine (54 118 cells) meshes at 8 L/s yields a difference of only 1.44%, with a Grid Convergence Index (GCI) of 0.60% on the fine grid, confirming the reliability of the numerical solution.

# 1 Introduction

The objective of this assignment is to investigate the pressure difference exerted on a bluff body object placed within a variable area channel. When fluid flows past an obstruction in a channel, boundary layer separation occurs at the edges of the body, forming a recirculation zone (wake) downstream. This produces a significant pressure difference between the upstream and downstream faces, which manifests as drag on the body.

Understanding pressure drop across obstructions is fundamental to many engineering applications, including flow measurement devices (orifice plates, Venturi meters), heat exchanger design, and structural loading analysis.

## 1.1 Aims and Objectives

- To simulate water flow at three distinct flow rates (4, 8, and 12 L/s) through a 2D channel containing a bluff body using the SST  $k-\omega$  turbulence model.
- To calculate and compare the pressure difference measured 25 mm upstream and 25 mm downstream of the top plate for each flow rate.
- To perform a grid independence study (verification) comparing coarse and fine meshes at the 8 L/s condition.
- To estimate numerical uncertainty using the Grid Convergence Index (GCI) methodology.
- To analyse the flow field using pressure contours, velocity contours, and streamline visualisations.

# 2 Simulation Methodology

## 2.1 Problem Definition and Geometry

The computational domain is a two-dimensional rectangular channel with the following dimensions:

- **Channel length:** 860 mm (0.86 m)
- **Channel height ( $H$ ):** 78 mm (0.078 m), spanning from  $y = -0.039$  m to  $y = 0.039$  m
- **Bluff body:** A plate attached to the top wall, spanning from  $x = 0.230$  m to  $x = 0.260$  m (30 mm wide), extending downward from the top wall to approximately  $y = 0.021$  m

The measurement locations for pressure extraction are:

- **Upstream line:**  $x = 0.205$  m (25 mm upstream of the plate leading edge)
- **Downstream line:**  $x = 0.285$  m (25 mm downstream of the plate trailing edge)

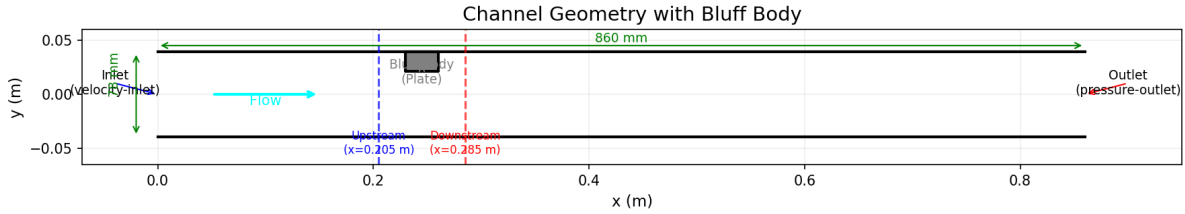


Figure 1: Schematic of the 2D channel geometry showing the bluff body plate, measurement lines, and boundary conditions.

## 2.2 Fluid Properties

Water is used as the working fluid with the properties listed in Table 1.

Table 1: Fluid properties (water at 20 °C).

| Property                    | Value    | Unit              |
|-----------------------------|----------|-------------------|
| Density ( $\rho$ )          | 998.2    | kg/m <sup>3</sup> |
| Dynamic viscosity ( $\mu$ ) | 0.001002 | Pa·s              |

## 2.3 Computational Grids

Two meshes were generated for this study, as summarised in Table 2. The fine mesh represents approximately a factor of four refinement in cell count over the coarse mesh, corresponding to a grid refinement ratio of  $r = \sqrt{54\,118/13\,580} = 1.996$  in two dimensions.

Table 2: Computational mesh details.

| Mesh   | Cells  | Nodes  | Faces   |
|--------|--------|--------|---------|
| Coarse | 13 580 | 11 963 | 25 542  |
| Fine   | 54 118 | 47 004 | 101 121 |

Both meshes were imported in GAMBIT format (.msh) with pre-defined boundary zones:

- Zone 6: velocity-inlet (inlet)
- Zone 5: pressure-outlet (outlet)

- Zone 4: wall (channel walls and bluff body surfaces)
- Zone 2: fluid (interior domain)

## 2.4 Boundary Conditions and Solver Setup

The simulations were performed in ANSYS Fluent 2024 R2 (2D, double precision, serial). The key solver settings are:

- **Solver:** Pressure-based, steady-state
- **Turbulence model:** SST  $k-\omega$  (Shear Stress Transport) [4], selected for its well-established accuracy in predicting boundary layer separation, adverse pressure gradient flows, and wake regions
- **Inlet:** Velocity-inlet with uniform velocity magnitude; turbulence specified via Intensity and Hydraulic Diameter method
- **Outlet:** Pressure-outlet at 0 Pa gauge pressure
- **Walls:** No-slip condition (default)
- **Spatial discretisation:** Second-order upwind
- **Initialisation:** Hybrid initialisation
- **Convergence:** 2000 maximum iterations (all cases converged well before this limit)

### 2.4.1 Hydraulic Diameter Calculation

For a rectangular channel (assuming unit depth  $W = 1$  m for the 2D formulation):

$$D_h = \frac{4A}{P} = \frac{4HW}{2(H + W)} \quad (1)$$

Substituting  $H = 0.078$  m and  $W = 1$  m:

$$D_h = \frac{4 \times 0.078 \times 1}{2(0.078 + 1)} = \frac{0.312}{2.156} = 0.1447 \text{ m} \quad (2)$$

### 2.4.2 Flow Condition Calculations

The inlet velocity is calculated from the volumetric flow rate:

$$V = \frac{Q}{A} = \frac{Q}{H \times W} \quad (3)$$

The Reynolds number is:

$$\text{Re} = \frac{\rho V D_h}{\mu} \quad (4)$$

The turbulence intensity is estimated using the empirical correlation for fully developed pipe flow:

$$I = 0.16 \times \text{Re}^{-1/8} \quad (5)$$

The resulting flow conditions for each case are summarised in Table 3.

Table 3: Flow condition calculations.

| Flow Rate (L/s) | Inlet Velocity (m/s) | Re     | TI (%) |
|-----------------|----------------------|--------|--------|
| 4               | 0.0513               | 7386   | 5.25   |
| 8               | 0.1026               | 14 771 | 4.82   |
| 12              | 0.1538               | 22 157 | 4.58   |

## 2.5 Simulation Cases

Four simulation cases were executed, as detailed in Table 4.

Table 4: Summary of simulation cases.

| Case ID    | Mesh                  | Flow Rate (L/s) | Purpose                         |
|------------|-----------------------|-----------------|---------------------------------|
| fine_4ls   | Fine (54 118 cells)   | 4               | Flow rate study                 |
| fine_8ls   | Fine (54 118 cells)   | 8               | Flow rate study + GIS reference |
| fine_12ls  | Fine (54 118 cells)   | 12              | Flow rate study                 |
| coarse_8ls | Coarse (13 580 cells) | 8               | Grid independence study         |

## 2.6 Post-Processing

Pressure differences were calculated by creating line surfaces at the upstream ( $x = 0.205$  m) and downstream ( $x = 0.285$  m) measurement locations, spanning the full channel height from  $y = -0.039$  m to  $y = 0.039$  m. The area-weighted average static pressure was computed on each line, and the pressure difference was obtained as:

$$\Delta P = P_{\text{upstream}} - P_{\text{downstream}} \quad (6)$$

### 3 Results and Discussion

#### 3.1 Convergence Behaviour

All four simulation cases achieved convergence before reaching the 2000-iteration limit, as shown in Table 5.

Table 5: Convergence summary.

| Case       | Iterations | Wall Time |
|------------|------------|-----------|
| fine_4ls   | 162        | ~68 s     |
| fine_8ls   | 151        | ~63 s     |
| fine_12ls  | 151        | ~60 s     |
| coarse_8ls | 146        | ~17 s     |

All residuals (continuity,  $x$ -velocity,  $y$ -velocity,  $k$ ,  $\omega$ ) dropped below the default convergence criteria ( $10^{-3}$ ). The rapid convergence confirms that the flow conditions are well-posed and the mesh quality is adequate.

Some reversed flow was observed at the pressure outlet during the initial iterations (up to ~27% of the outlet area for the higher flow rates), which is expected for bluff body flows where wake recirculation can extend to the outlet boundary. This reversed flow diminished as the solution converged.

#### 3.2 Pressure Difference Analysis (Fine Grid)

The area-weighted average static pressures at the upstream and downstream measurement lines and the resulting pressure differences are presented in Table 6.

Table 6: Pressure difference results (fine grid).

| Flow Rate (L/s) | Re     | $P_{\text{up}}$ (Pa) | $P_{\text{down}}$ (Pa) | $\Delta P$ (Pa) |
|-----------------|--------|----------------------|------------------------|-----------------|
| 4               | 7386   | 69.53                | -14.07                 | 83.60           |
| 8               | 14 771 | 276.92               | -56.30                 | 333.22          |
| 12              | 22 157 | 619.26               | -126.16                | 745.42          |

Several observations can be drawn from these results:

1. **Positive upstream pressure:** The static pressure upstream of the bluff body is positive (above the outlet reference of 0 Pa) for all flow rates, reflecting the stagnation effect as the flow approaches the obstruction.

2. **Negative downstream pressure:** The downstream pressure is consistently negative, indicating a low-pressure wake region behind the plate. This is characteristic of bluff body flows where boundary layer separation creates a recirculation zone.
3. **Increasing pressure drop with flow rate:** The pressure difference increases significantly with flow rate, from 83.60 Pa at 4 L/s to 745.42 Pa at 12 L/s—nearly a factor of nine increase for a factor of three increase in flow rate.

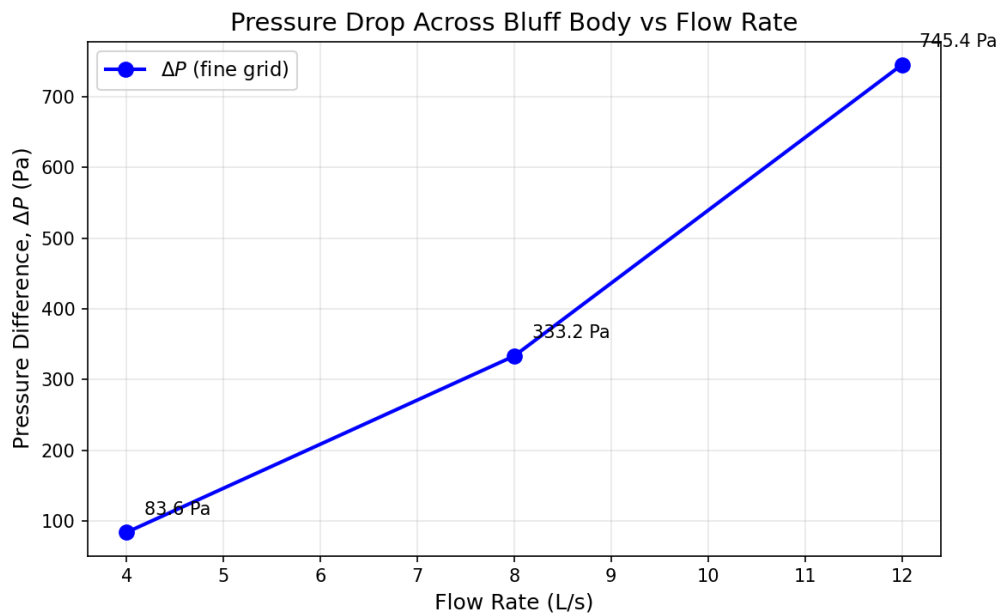


Figure 2: Pressure difference across the bluff body as a function of flow rate (fine grid).

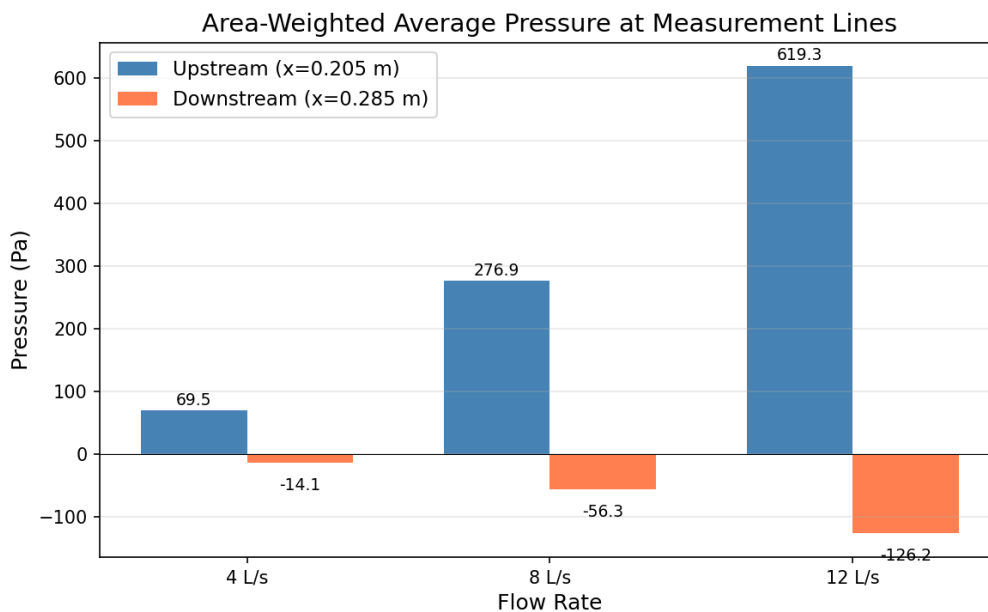


Figure 3: Area-weighted average pressures at the upstream and downstream measurement lines.

### 3.2.1 Relationship Between Pressure Drop and Velocity

For flow past a bluff body, the pressure drag is expected to be proportional to the dynamic pressure, which scales as  $V^2$ :

$$\Delta P \propto \frac{1}{2} \rho V^2 \quad (7)$$

Plotting the pressure difference against  $V^2$  (Figure 4) reveals a near-linear relationship, confirming that the dominant mechanism is form (pressure) drag rather than skin friction. The linear fit yields:

$$\Delta P \approx 31\,497 V^2 + 0.9 \text{ Pa} \quad (8)$$

from which an approximate pressure loss coefficient can be estimated:

$$K = \frac{\Delta P}{\frac{1}{2} \rho V^2} = \frac{2 \times 31\,497}{998.2} \approx 63.1 \quad (9)$$

This high loss coefficient is consistent with a bluff body obstruction in a confined channel, where the blockage ratio significantly amplifies the pressure drop compared to an unconfined body.

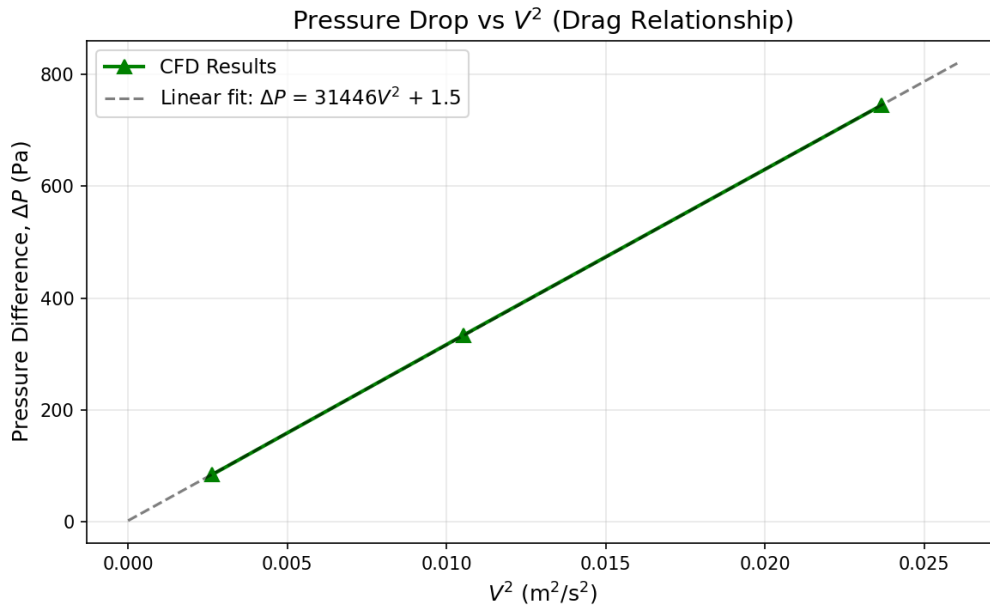


Figure 4: Pressure difference vs.  $V^2$ , demonstrating the expected quadratic relationship.

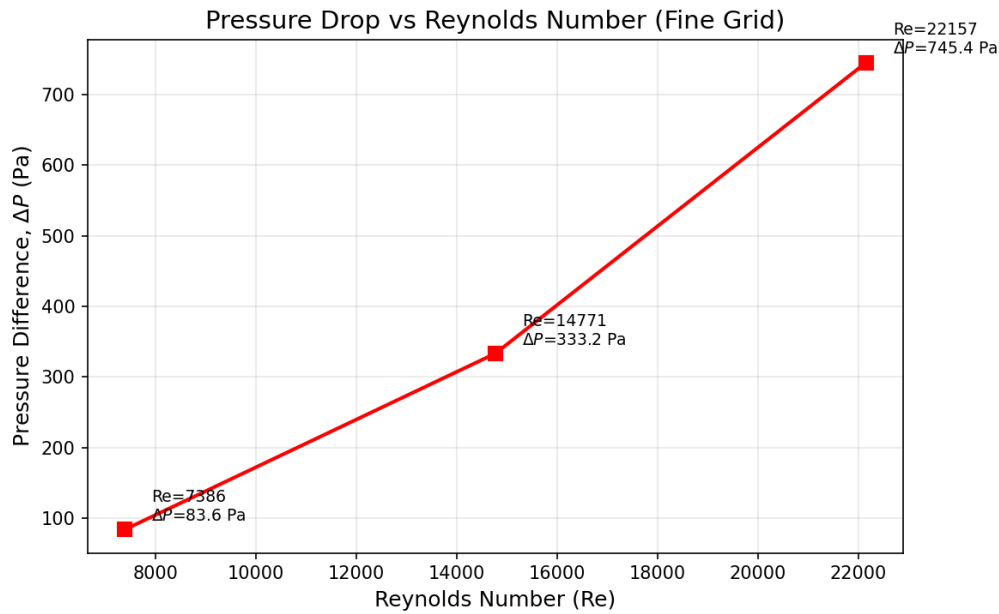


Figure 5: Pressure difference vs. Reynolds number.

### 3.3 Verification and Validation (V&V)

#### 3.3.1 Grid Independence Study (8 L/s)

A grid independence study was performed at 8 L/s by comparing results from the coarse (13 580 cells) and fine (54 118 cells) meshes. The results are presented in Table 7.

Table 7: Grid independence study results (8 L/s).

| Mesh   | Cells  | $\Delta P$ (Pa) | % Difference |
|--------|--------|-----------------|--------------|
| Coarse | 13 580 | 328.41          | —            |
| Fine   | 54 118 | 333.22          | 1.44%        |

The percentage difference between the coarse and fine mesh solutions is only 1.44%, indicating that the solutions are approaching grid independence. The coarse mesh slightly under-predicts the pressure difference compared to the fine mesh.

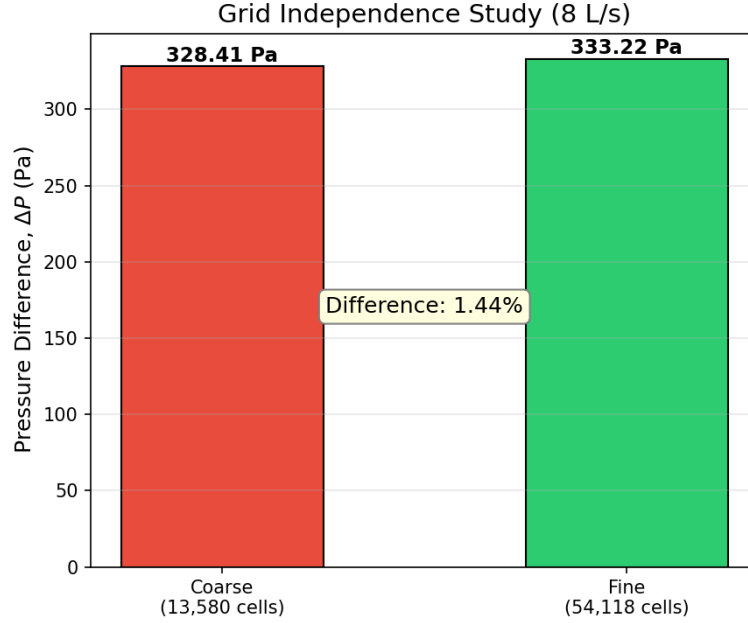


Figure 6: Grid independence study comparing coarse and fine mesh results at 8 L/s.

### 3.3.2 Grid Convergence Index (GCI)

The Grid Convergence Index (GCI) provides a formal estimate of numerical uncertainty due to spatial discretisation [6, 3]. The GCI is calculated as follows.

**Grid refinement ratio:**

$$r = \left( \frac{N_{\text{fine}}}{N_{\text{coarse}}} \right)^{1/d} = \left( \frac{54\,118}{13\,580} \right)^{1/2} = 1.996 \quad (10)$$

where  $d = 2$  is the spatial dimension.

**Observed order of convergence:** With only two grid levels available, the formal order  $p$  cannot be computed directly. The theoretical second-order accuracy of the discretisation scheme is assumed ( $p = 2$ ).

**Relative error:**

$$\varepsilon = \frac{|f_{\text{fine}} - f_{\text{coarse}}|}{f_{\text{fine}}} = \frac{|333.22 - 328.41|}{333.22} = 0.01444 \quad (1.44\%) \quad (11)$$

**GCI on the fine grid** (with safety factor  $F_s = 1.25$ ):

$$\text{GCI}_{\text{fine}} = \frac{F_s \cdot |\varepsilon|}{r^p - 1} = \frac{1.25 \times 0.01444}{1.996^2 - 1} = 0.00605 \quad (0.60\%) \quad (12)$$

**Richardson extrapolation estimate:**

$$f_{\text{exact}} \approx f_{\text{fine}} + \frac{f_{\text{fine}} - f_{\text{coarse}}}{r^p - 1} = 333.22 + \frac{333.22 - 328.41}{1.996^2 - 1} = 334.83 \text{ Pa} \quad (13)$$

The GCI of 0.60% on the fine grid indicates a high level of confidence in the numerical solution. The fine grid solution of 333.22 Pa is within  $\pm 0.60\%$  of the grid-independent value, estimated at approximately 334.83 Pa by Richardson extrapolation.

The small difference between coarse and fine mesh results (1.44%) and the low GCI (0.60%) confirm that the fine mesh provides a grid-independent solution for the pressure difference. The coarse mesh, despite having approximately one-quarter the number of cells, still captures the overall flow physics accurately, though with slightly reduced precision in the wake region where gradients are steepest.

For a more rigorous GCI analysis, a third grid level would be needed to verify the observed order of convergence. Additionally, the GCI methodology assumes that the solution is in the asymptotic range of convergence, which the low error values suggest is the case here.

**3.4 Flow Field Analysis****3.4.1 Velocity Profiles**

The  $x$ -direction velocity profiles extracted at the upstream ( $x = 0.205 \text{ m}$ ) and downstream ( $x = 0.285 \text{ m}$ ) measurement lines provide detailed insight into the flow behaviour around the bluff body.

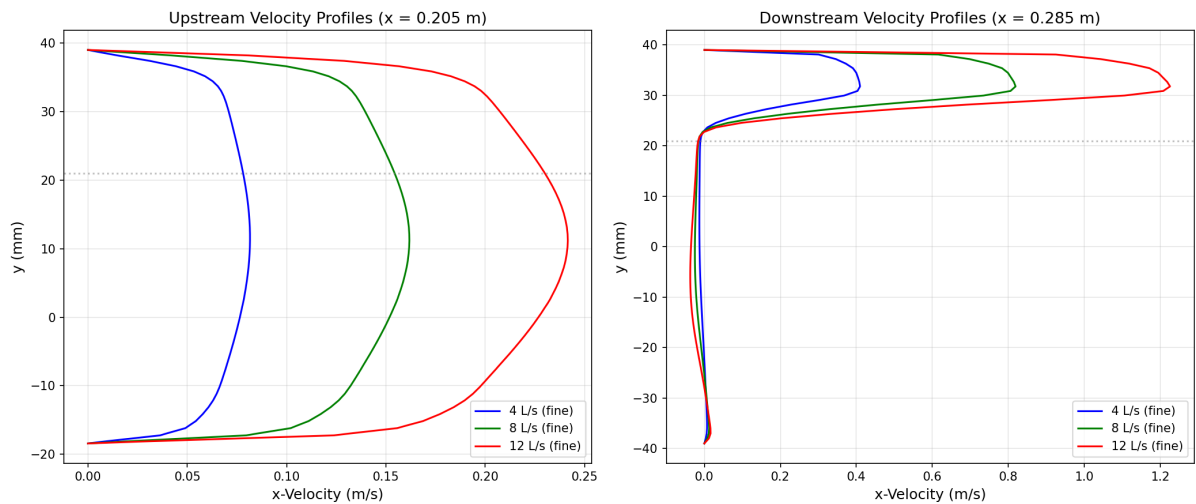


Figure 7: Comparison of  $x$ -velocity profiles at upstream and downstream measurement lines for all three flow rates (fine grid).

**Upstream profiles ( $x = 0.205$  m):** The upstream velocity profiles show a smooth, approximately parabolic shape characteristic of developing channel flow. The profiles are symmetric about the channel centreline. Peak velocities are approximately 0.082 m/s, 0.162 m/s, and 0.242 m/s for the 4, 8, and 12 L/s cases respectively. These peak velocities are approximately 1.6 times the mean inlet velocity, consistent with the velocity distribution in a developing turbulent channel flow.

**Downstream profiles ( $x = 0.285$  m):** The downstream velocity profiles reveal the wake structure behind the plate. Key observations include:

- **Velocity deficit in the wake:** Near the top wall ( $y > 21$  mm, behind the plate), the velocity drops significantly. The flow in this region shows either very low positive velocity or negative velocity (reverse flow), confirming the existence of a recirculation zone in the plate wake.
- **Flow acceleration below the plate:** Below the plate ( $y < 21$  mm), the velocity is significantly higher than the upstream value. At 8 L/s, the peak downstream velocity reaches approximately 0.82 m/s compared to the upstream peak of 0.16 m/s—a factor of five increase. This acceleration occurs because the flow must pass through the reduced cross-section between the plate bottom and the lower wall.
- **Negative velocities:** Small regions of negative  $x$ -velocity (up to  $-0.025$  m/s at 8 L/s) are observed in the wake, confirming boundary layer separation and recirculation.

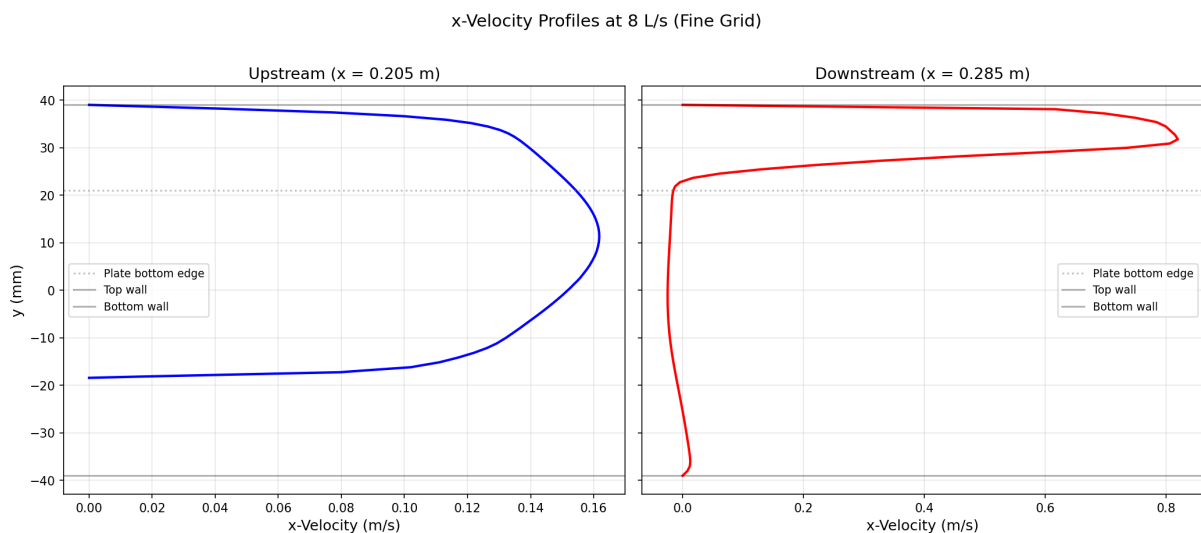


Figure 8: Detailed upstream and downstream velocity profiles for the 8 L/s case (fine grid), showing the dramatic flow redistribution caused by the bluff body.

### 3.4.2 Grid Independence of Velocity Profiles

The velocity profiles from the fine and coarse meshes at 8 L/s are compared in Figure 9.

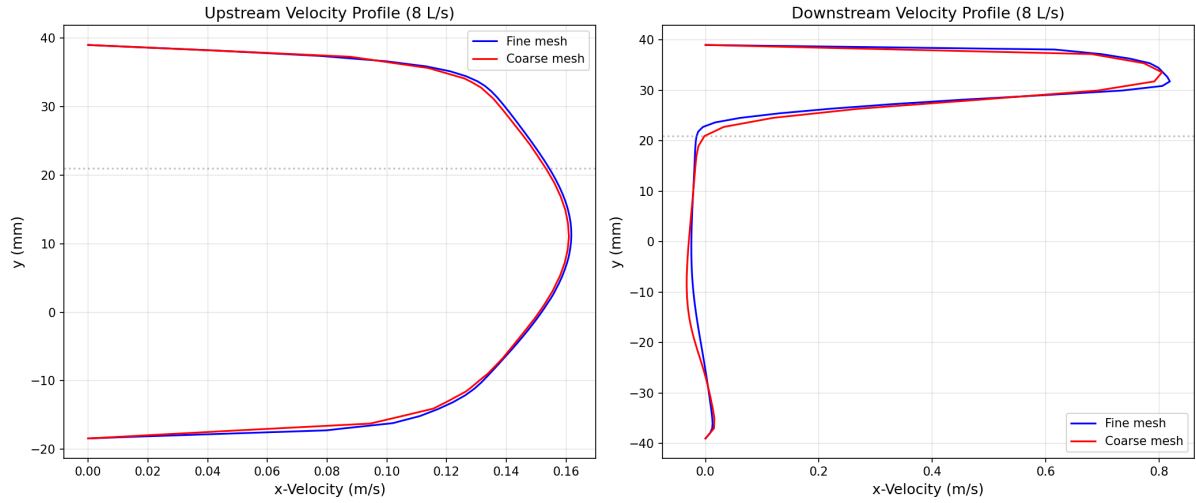


Figure 9: Comparison of velocity profiles between fine and coarse meshes at 8 L/s.

The coarse and fine mesh velocity profiles show excellent agreement at the upstream location, where flow gradients are relatively mild. At the downstream location, the profiles also agree well, with minor differences in the wake region where the coarse mesh has fewer cells to resolve the steep velocity gradients. This is consistent with the 1.44% difference observed in the pressure drop values.

### 3.4.3 Pressure Profiles

The static pressure distributions along the measurement lines (Figure 10) show the pressure variation across the channel height.

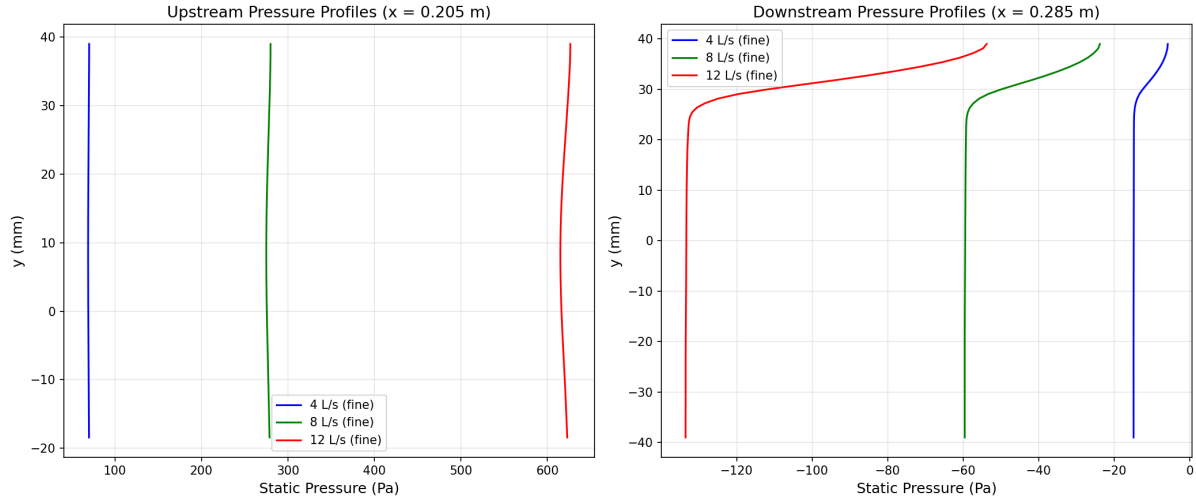


Figure 10: Static pressure profiles at upstream and downstream measurement lines for all three flow rates.

The upstream pressure profiles are relatively uniform across the channel height, indicating that the flow is not yet significantly influenced by the plate at  $x = 0.205$  m (25 mm upstream of the plate leading edge). The downstream pressure profiles show more variation, reflecting the complex wake structure.

### 3.4.4 Flow Physics Discussion

The flow past the bluff body plate exhibits several important fluid dynamic phenomena:

1. **Approach flow:** The flow enters the channel as a uniform stream and develops a boundary layer along the channel walls. At the upstream measurement line, the velocity profile reflects the developing boundary layer.
2. **Flow acceleration:** As the flow approaches the plate, it must accelerate to pass through the reduced cross-sectional area between the bottom of the plate and the lower channel wall. This is confirmed by the significantly higher velocities observed in the downstream profiles below  $y = 21$  mm.
3. **Boundary layer separation:** The flow separates at the edges of the plate, forming shear layers that bound a recirculation zone in the wake. The negative velocities in the downstream profiles confirm this separation.
4. **Wake region:** Downstream of the plate, a low-pressure recirculation zone forms. The downstream measurement line captures this wake region, which explains both the negative static pressure and the velocity deficit observed.
5. **Pressure recovery:** Further downstream, the wake gradually closes as the separated shear layers merge and the pressure recovers toward the outlet value.

### 3.4.5 Contour Visualisation

The saved case files (.cas.h5 and .dat.h5) for all four simulation cases can be opened in the ANSYS Fluent GUI to produce contour visualisations:

- **Pressure contours:** Filled contours of Static Pressure reveal the high-pressure stagnation region upstream and the low-pressure wake downstream of the plate.
- **Velocity magnitude contours:** Filled contours of Velocity Magnitude identify the flow acceleration around the plate and the low-velocity wake region.
- **Streamlines:** Streamlines from the inlet visualise the flow pattern, including the recirculation zone behind the plate.

## 4 Conclusion

This CFD study has successfully investigated the pressure difference across a bluff body (plate) in a two-dimensional rectangular channel at three flow rates using ANSYS Fluent with the SST  $k$ - $\omega$  turbulence model.

### Key Findings

1. **Pressure drop results:** The pressure differences measured 25 mm upstream and downstream of the plate are:
  - 4 L/s ( $Re = 7\,386$ ):  $\Delta P = 83.60$  Pa
  - 8 L/s ( $Re = 14\,771$ ):  $\Delta P = 333.22$  Pa
  - 12 L/s ( $Re = 22\,157$ ):  $\Delta P = 745.42$  Pa
2. **Quadratic relationship:** The pressure drop exhibits a near-quadratic relationship with flow velocity ( $\Delta P \propto V^2$ ), confirming that form drag is the dominant pressure loss mechanism, as expected for a bluff body in a confined channel.
3. **Grid independence:** The grid independence study at 8 L/s demonstrates excellent convergence between the coarse (328.41 Pa) and fine (333.22 Pa) mesh solutions, with a difference of only 1.44%.
4. **Numerical uncertainty:** The Grid Convergence Index (GCI) on the fine grid is 0.60%, indicating that the numerical solution is well within the asymptotic convergence range and the discretisation error is small.
5. **Solver convergence:** All cases converged within 146–162 iterations (well below the 2000-iteration limit), confirming well-posed problem formulation and adequate mesh quality.

## Limitations and Recommendations

- The simulation uses a 2D approximation, which does not capture three-dimensional effects such as end-wall boundary layers and horseshoe vortices.
- A uniform inlet velocity profile was used; a  $\frac{1}{7}$ th power law profile would provide a more realistic representation of fully developed turbulent channel flow.
- Only two grid levels were available for the GCI analysis. A three-grid study would provide a more rigorous estimate of the observed order of convergence.
- The SST  $k-\omega$  model, while appropriate for wall-bounded flows with separation, is a RANS model that provides time-averaged results. For more detailed analysis of wake dynamics, Large Eddy Simulation (LES) could be considered.

## References

## References

- [1] ANSYS Inc. (2024). *ANSYS Fluent Theory Guide*, Release 2024 R2.
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- [3] Celik, I.B., Ghia, U., Roache, P.J., Freitas, C.J., Coleman, H. and Raad, P.E. (2008). "Procedure for Estimation and Reporting of Uncertainty Due to Discretization in CFD Applications." *Journal of Fluids Engineering*, 130(7), 078001.
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- [5] Rana, Z. (2024). *Verification & Validation for Industrial Applications*. Cranfield University lecture notes.
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## A Simulation Setup Parameters

### A.1 Solver Settings

Table 8: ANSYS Fluent solver settings.

| Parameter                  | Value                             |
|----------------------------|-----------------------------------|
| Solver type                | Pressure-based                    |
| Formulation                | Steady-state                      |
| Dimension                  | 2D                                |
| Precision                  | Double                            |
| Turbulence model           | SST $k-\omega$                    |
| Near-wall treatment        | Default (automatic wall function) |
| Spatial discretisation     | Second-order upwind               |
| Pressure-velocity coupling | SIMPLE                            |
| Initialisation             | Hybrid                            |
| Max iterations             | 2000                              |

### A.2 Boundary Conditions

Table 9: Boundary condition summary.

| Boundary | Type            | Parameters                                                           |
|----------|-----------------|----------------------------------------------------------------------|
| Inlet    | Velocity-inlet  | $V = f(Q)$ , $I = 0.16 \text{ Re}^{-1/8}$ , $D_h = 0.1447 \text{ m}$ |
| Outlet   | Pressure-outlet | $P_{\text{gauge}} = 0 \text{ Pa}$                                    |
| Walls    | No-slip wall    | Stationary, adiabatic                                                |

## B Complete Results Data

### B.1 Pressure Results

Table 10: Complete pressure results for all simulation cases.

| Case       | Mesh   | $Q$<br>(L/s) | $V$<br>(m/s) | Re     | TI<br>(%) | $P_{\text{up}}$<br>(Pa) | $P_{\text{down}}$<br>(Pa) | $\Delta P$<br>(Pa) |
|------------|--------|--------------|--------------|--------|-----------|-------------------------|---------------------------|--------------------|
| fine_4ls   | Fine   | 4            | 0.0513       | 7 386  | 5.25      | 69.53                   | -14.07                    | 83.60              |
| fine_8ls   | Fine   | 8            | 0.1026       | 14 771 | 4.82      | 276.92                  | -56.30                    | 333.22             |
| fine_12ls  | Fine   | 12           | 0.1538       | 22 157 | 4.58      | 619.26                  | -126.16                   | 745.42             |
| coarse_8ls | Coarse | 8            | 0.1026       | 14 771 | 4.82      | 273.01                  | -55.40                    | 328.41             |

## B.2 Grid Convergence Index

Table 11: Grid Convergence Index parameters and results.

| Parameter                | Value        |
|--------------------------|--------------|
| $N_{\text{coarse}}$      | 13 580 cells |
| $N_{\text{fine}}$        | 54 118 cells |
| Refinement ratio ( $r$ ) | 1.996        |
| Assumed order ( $p$ )    | 2            |
| $f_{\text{coarse}}$      | 328.41 Pa    |
| $f_{\text{fine}}$        | 333.22 Pa    |
| Relative error           | 1.44%        |
| $GCI_{\text{fine}}$      | 0.60%        |
| $GCI_{\text{coarse}}$    | 2.41%        |
| Richardson extrapolation | 334.83 Pa    |